

25PY101: Engineering Physics

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Scheme of evaluation

M2T2 Review

Question-to-topic mapping

The topic related to question asked in part-A of M2T1 assessment is given below.

Question	Topic
Q1	QFET
Q2	Opto-electronics
Q3	Optical fibres and Lasers
Q4	Quantum mechanics principles
Q5	Quantum mechanics principles
Q6	Lasers and Optical fibres
Q7	QFET
Q8	Lasers and Optical fibres
Q9	QBT
Q10	Opto-electronics

T2 review protocol

The T2 review consists of two rounds of viva:

1. Intermediate Level Viva
2. Advanced Level Viva

Each round may be answered either:

- Individually
- Group-wise (with an elected group leader)

As a batch, you have to make **two** decisions –

- Answer intermediate level viva as group or individually
- Answer advanced level viva as group or individually...

Intermediate level viva

Rules

- Maximum initial marks allotted = 6
- Marks are deducted for wrong answers
- Individual mode:
 - Number of questions = 3
 - Deduction per wrong answer = -2
- Group mode:
 - Number of questions = 1
 - Deduction per wrong answer = -4
- Group leader will get $+1$.

Possible scores

(A) Individual ode

Let w be the number of wrong answers.

$$\text{Marks} = 6 - 2w$$

Since there are 3 questions:

$$w \in \{0, 1, 2, 3\}$$

Wrong Answers	Final Marks
0	6
1	4
2	2
3	0

Thus, the possible scores are:

$$\boxed{\{0, 2, 4, 6\}}$$

(B) Group mode

Only one question is asked.

$$\text{Marks} = \begin{cases} 6, & \text{if correct} \\ 2, & \text{if wrong} \end{cases}$$

Outcome	Final Marks
Correct	6
Wrong	2

Thus, the possible scores are:

$$\{2, 6\}$$

Advanced level viva

Rules

- Maximum marks = 4
- Marks are awarded for correct answers
- Individual mode:
 - Number of questions = 2
 - Marks per correct answer = +2
- Group mode:
 - Number of questions = 1
 - Marks for correct answer = +4
- Group leader will get +1.

Possible scores

(A) Individual mode

Let c be the number of correct answers.

$$\text{Marks} = 2c$$

Since there are 2 questions:

$$c \in \{0, 1, 2\}$$

Correct Answers	Final Marks
0	0
1	2
2	4

Thus, the possible scores are:

$$\boxed{\{0, 2, 4\}}$$

(B) Group mode

Only one question is asked.

$$\text{Marks} = \begin{cases} 4, & \text{if correct} \\ 0, & \text{if wrong} \end{cases}$$

Outcome	Final Marks
Correct	4
Wrong	0

Thus, the possible scores are:

$$\boxed{\{0, 4\}}$$

Overall possible total scores

Total marks are obtained by combining:

- Intermediate Viva marks
- Advanced Viva marks

(A) Individual in both rounds

Possible totals:

$$\{0, 2, 4, 6\} + \{0, 2, 4\}$$

Hence,

$$\boxed{\{0, 2, 4, 6, 8, 10\}}$$

(B) Group in both rounds

Possible totals:

$$\{2, 6\} + \{0, 4\}$$

Hence,

$$\boxed{\{2, 6, 10\}}$$

(C) Mixed possibilities

- Intermediate Individual + Advanced Group

$$\{0, 2, 4, 6\} + \{0, 4\} = \boxed{\{0, 2, 4, 6, 8, 10\}}$$

- Intermediate Group + Advanced Individual

$$\{2, 6\} + \{0, 2, 4\} = \boxed{\{2, 4, 6, 8, 10\}}$$

End of Scheme